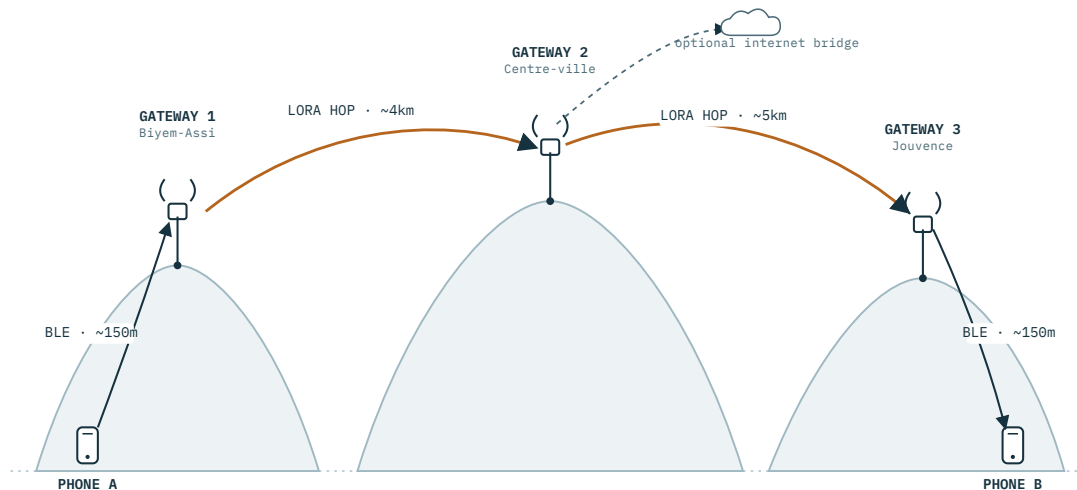


The LoRa Gateway Mesh

How a text message or voice note gets from one Yaoundé neighborhood to another with **no SIM card, no mobile internet, and no airtime credit** — relayed hilltop to hilltop over long-range radio instead of a cell network.

The physical path

Bluetooth only reaches ~150m, so it can't cross a city on its own. LoRa radio bridges the gap between gateway nodes mounted on high ground — Yaoundé's hills work in the network's favor here.



A message hops Bluetooth → LoRa → LoRa → Bluetooth across three hilltop gateways, arcing above the terrain that blocks a direct line of sight. The payload is encrypted end-to-end — gateways relay it but can't read it — and no cellular tower is involved at any point.

Adding a contact

There's no phone directory to search — every contact starts as a deliberate exchange of a public key. Three ways that exchange happens, all zero-backend:

IN PERSON

Live QR Scan

Open your profile QR, they scan it with their camera. Instant, no channel needed — the gold standard for knowing you added the right person.

SHARED IMAGE

Screenshot & Send

Send your QR as an image through any channel you already have — WhatsApp, SMS, email. They scan it from their gallery instead of live. Some apps recompress images enough to break the scan.

SHARED LINK

Text Code / Deep Link

A plain-text code or link carrying the same key. Survives being forwarded through any text channel without the image-compression risk.

NEARBY, OPT-IN

Meeting Someone New

A Bluetooth presence beacon shows "people running the app near you right now" — proximity only, not a citywide search. Off by default; visibility should be a choice, not passive broadcasting.

Two things this creates are easy to conflate, so it's worth separating them explicitly:

EPHEMERAL

Nearby List

Shows only while both phones are within Bluetooth range of each other — disappears the moment either one walks away. Answers "who's physically near me right now."

PERSISTENT

Saved Contact

Once added — by any method above — stays in your contact list forever. Lets you message them anytime, anywhere, routed through the mesh. No live "online" status; delivery depends on gateway coverage between you.

What happens to one message

1

Compose & encrypt. Phone A encrypts the message to the recipient's public key — only the destination ID travels unencrypted.

2

Hand off over Bluetooth. The phone connects to whichever gateway is currently in range. No pairing account, no SIM.

3

Flood across the mesh. Gateways relay the packet over LoRa, hop to hop, with a TTL and dedupe ID, until one of them has the recipient's phone nearby.

4

Buffer if offline. If Phone B isn't connected anywhere yet, its nearest gateway holds the message — typically 24 to 72 hours — and keeps listening.

5

Deliver over Bluetooth. The moment Phone B connects to any gateway on the network, the message is handed over and decrypted locally.

6

Optional acknowledgment. A delivery receipt can flow back along the same path — the mesh equivalent of a "seen" tick.

Four layers

CLIENT

Phone App

React Native app on every user's phone. Generates a keypair on first install, compresses voice notes (Opus, ~6–8kbps), and queues messages locally when no gateway is in range.

ACCESS

Gateway Nodes

ESP32 + LoRa radio + Bluetooth, solar-powered, mounted on hilltops and rooftops. The only hardware end users ever touch — over Bluetooth, from up to ~150m away.

BACKBONE

LoRa Mesh

Gateways relay to each other over 868MHz radio, flooding packets until one of them has the recipient connected. Yaoundé's hills extend each hop's reach.

OPTIONAL

Hybrid Bridge

Any gateway with real internet — an office WiFi, a partner site — can relay across it instead of waiting for a radio path. Speeds up delivery; never required of the end users.

Gateway node — bill of materials

COMPONENT	ROLE	APPROX. COST
LoRa radio board (e.g. Heltec LoRa32 V3)	Long-range relay + Bluetooth peripheral	\$20–35
Tuned antenna, elevated mount	Extends hop range from hilltop sites	~\$10
Solar panel + battery	Off-grid power	\$15–25
Weatherproof enclosure	Outdoor mounting	~\$10
Per node, one-time		~\$55–80 · no monthly fee

Where the line actually is

Works today

- ✓ Text messages, city-wide, no SIM or internet
- ✓ Voice notes — delivery in seconds to minutes
- ✓ Instant delivery within direct Bluetooth range (no relay)

Not possible with this design

- ✗ Live, real-time phone calls beyond direct range
- ✗ Live video calls
- ✗ Guaranteed delivery time — depends on gateway density

Distribution: Android first — the installer spreads phone to phone via APK sharing once seeded, no internet needed after the first copy. iOS needs an App Store touch per device.

Regulatory: LoRa runs on license-exempt 868MHz spectrum, but a city-scale gateway deployment should be confirmed with Cameroon's ART before installation.